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Review

Muscle activity and scapular kinematics in individuals with multidirectional shoulder instability: A systematic review

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ABSTRACT

Background: Several studies reported the importance of glenohumeral and scapular muscle activity and scapular kinematics in multidirectional shoulder instability (MDI), yet a systematic overview is currently lacking.
Objective: This systematic review evaluates and summarizes the evidence regarding muscle activity and shoulder kinematics in individuals with MDI compared to healthy controls.
Method: The electronic databases PubMed and Web of Science were searched in September 2020 with key words regarding MDI (population), muscle activity, and glenohumeral and scapular movement patterns (outcomes). All studies that compared muscle activity or scapular kinematics between shoulders with MDI and healthy shoulders were eligible for this review, except for case reports and case series. All articles were screened on the title and abstract, and remaining eligible articles were screened on full text. The risk of bias of included articles was assessed by a checklist for case–control data, as advised by the Cochrane collaboration.
Results: After full text screening, 12 articles remained for inclusion and one study was obtained by hand search. According to the guidelines of the Dutch Institute for Healthcare Improvement, most studies were of moderate methodological quality. We found moderate evidence that MDI individuals show increased or prolonged activity of several rotator cuff muscles that control and centre the humeral head. Furthermore, we found evidence of decreased and/or shortened activity of muscles that move or accelerate the arm and shoulder girdle as well as increased and/or lengthened activity of muscles that decelerate the arm and shoulder girdle. The most consistent kinematic finding was that MDI individuals show significantly less upward rotation and more internal rotation of the scapula during elevation of the arm in the scapular plane as compared with controls. Finally, several studies also suggest that the humeral head demonstrates increased translations relative to the glenoid surface.
Conclusion: There is moderate evidence for altered muscle activity and altered humeral and scapular kinematics in MDI individuals as compared with controls.
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1. Introduction

Multidirectional shoulder instability (MDI) is a symptomatic condition of the glenohumeral joint, in which the humeral head cannot be maintained within the glenoid fossa [1,2], which leads to recurrent involuntary shoulder (sub)luxations. MDI is associated with symptoms ranging from discomfort and apprehension to chronic shoulder pain [3–6]. Since Neer and Foster (1980) [7] first described the concept, many authors have defined MDI as "symptomatic glenohumeral joint instability in more than one direction" [8–10]. However, this definition is general and rather vague [11–13].

Shoulder stability is a complex interplay between static (articular) and dynamic (mostly extra-articular) stabilizers [8,14], of which the relative contribution differs according to the position of the shoulder joint [15,16]. Static stabilizers, such as the glenoid labrum, shape of the glenoid surface and humeral head, intra-articular suction force, and tension of the shoulder capsule all limit excessive glenohumeral translations [15–18]. Capsular redundancy appears to be one of the factors that considerably contribute to MDI [19] because it is a risk

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factor for larger glenohumeral translations [2,5,19]. By illustration, recent studies in MDI have reported inferior humeral translations as large as 0.6 to 2.1 mm [20,21].

Furthermore, during mid-range positions and movements, dynamic stabilizers, such as the rotator cuff and scapulothoracic muscles, have a crucial role in shoulder stability. Because of their close proximity to the joint, rotator cuff muscles generate compressive forces, which pull the humeral head into the glenoid and increase stability [2,22]. For instance, during flexion and extension movements, anteroposterior translations of the humeral head are restricted by rotator cuff muscles that are activated in a direction-specific manner to ensure joint stability [23]. Besides the rotator cuff, the scapular muscles are important in maintaining glenohumeral stability [6,8,18]. Sufficient and timely activation is required to ensure an optimal scapular position during arm movements [18]. Previous research has shown that as much as 57% to 80% of MDI individuals present aberrant scapular movement patterns [18]. This scapular dyskinesis may have multiple possible causes, including insufficient strength or muscle length [18], inadequate proprioception [8], altered movement patterns to avoid pain [22], or sport-specific alterations [24]. Conversely, scapular dyskinesis may also occur as a result of shoulder instability [25].

Regardless of its cause, scapular dyskinesis has a strong effect on the inclination of the glenoid and the amount of subacromial space. As such, it is a risk factor for increased humeral translations and increased loads onto static stabilizing structures, including the labrum and joint capsule [8,12,17,18,25,26]. For instance, increased scapular protraction may generate much larger tensile loads on the anterior inferior glenohumeral ligament and thereby jeopardize shoulder stability in individuals with an existing ligamentary laxity [27]. Furthermore, insufficient scapular upward rotation implies less upward inclination of the glenoid cavity, which facilitates caudal translation or even subluxation of the humeral head [28].

Taken together, several studies reported the importance of glenohumeral and scapular muscle activity and scapular kinematics in MDI, yet a systematic overview is currently lacking. The present review summarizes the existing literature to gain insight into the mechanisms contributing to MDI, to assist clinicians in treatment and guide further research.

2. Methods

2.1. Information sources

This systematic review was conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines [29]. Electronic databases PubMed and Web of Science were searched in September 2020 to identify relevant articles. Two researchers (VS, predoctoral researcher, and IDW, postdoctoral researcher), working in the field of shoulder instability and trained in conducting systematic reviews, screened all articles to assess eligibility for inclusion. Articles were not excluded based on publication date.

2.2. Eligibility criteria and search strategy

The research question was defined using the Patient-Intervention-Comparison-Outcome-Study (PICOS) approach. Articles were included if they provided results of clinical studies (S) that evaluated muscle activity, scapular motion, or scapular position (O) in MDI individuals (P). The intervention (I) and comparison (C) were not defined. The search strategy was based on a combination of keywords and MeSH terms (Table 1) derived from the PICOS question.

2.3. Study selection

Studies were included if they met the following inclusion criteria:

- adult participants ($\geq$ 18 years old) with MDI;
- assessed muscle activity, scapular motion or scapular position;
- the study design was a randomized or non-randomized controlled trial, case–control study, cross-sectional, or cohort study;
- full texts were available in English or Dutch;
- MDI indviduals were compared with healthy controls.

First, articles were screened based on the title and abstract with the selection criteria above, then full texts of selected articles were evaluated for eligibility.

2.4. Data items and collection

Information was extracted and summarized in the evidence table (Table 2). The following data were included:

- author and year of publication;
- characteristics of the study population;
- research protocol;
- primary outcome measures;
- main findings.

2.5. Quality assessment: risk of bias in individual studies

Methodological quality of included articles was assessed by the checklist for case–control studies provided by the Dutch Institute for Healthcare Improvement (CBO) and the Dutch Cochrane Centre (www.cbo.nl/en). Only information regarding the comparison between MDI individuals (before any treatment occurred) and healthy controls was extracted. As such, from the studies of Illyés et al. [30], Kiss et al. [31] and Nyiri et al. [32], only pre-treatment

Table 1
Key words and MeSH terms in the PICO question.

Patients		Outcomes	
Multidirectional shoulder instability	MDI Shoulder/glenohumeral multidirectional	Muscle activity pattern	Muscle activity/activation/timing strength/force/balance "electromyography" [MeSH]
	"Shoulder dislocation"[MeSH] Disloc*/unstab*/sublux* "joint instability" [MeSH] Hyperextensib*/hyperlaxity/hypermob* laxity "Ehlers-Danlos/Ehlers-Danlos syndrome"[MeSH]	Movement pattern	Scapula Position/rhythm/control movement/motion/mobility Kinemat* Biomech* "Imaging, three-dimensional" [MeSH] "Biomechanical phenomena" [MeSH]

Table 2
The evidence table.

Author (year)	Study design	Study population	Research protocol		Main outcome measures	Main results
			Measurements	Task		
Barden et al. (2005) [35]	Case control	P: 7 ♂ MDI C: 11 ♂ normal, healthy shoulders	Double-differential surface electrodes on: -AD, MD, PD -LD -PM pars Sternalis Bipolar intramuscular fine wire electrodes: -on -IS -SS	Shoulder movements on KinCom 500H isokinetic dynamometer: -Abd/add (10° → 100°)]: lying on the side -flexion/extension (20° → 110°]: seated -IR/ER (20° → 110°]: seated with the arm in 90° abduction and 90° elbow flexion Modalities: 3 trials × 10 consecutive isotonic reps of 35 N at 170°/sec	Four temporal EMG parameters: -onset of activation (ONST) -termination of activation (TERM) -normalized average amplitude (AMP) -duration of activation in terms of range of motion (ACT)	SS: -AMP, ONST, TERM: MDI = C -ER ACT: MDI < C IS: -AMP: MDI = C -ER ACT: MDI < C -PD: -ONST ER: delayed in MDI -ER ACT approached significance at $P < 0.09$ due to -ONST PM: -AMP extension: MDI < C -ONST: MDI = C -TERM, ACT extension: MDI terminated later, active throughout an extended range of motion
Illyés and Kiss (2007) [37]	Case control	P: 15 MDI; 10 ♀ 5 ♂ C: 15 normal, healthy shoulders; 10 ♀ 5 ♂	ZEBRIS surface EMG: Monopolar surface electrodes on: -PM -IS -AD, MD, PD -UT -BB -TB	(1) 3 isokinetic movements: -pull in sagittal plane -forward punch in sagittal plane -elevation in scapular plane (2) 3 dynamic movements: -slow overhead throw -rapid overhead throw at maximal speed Modalities: (1) 3 reps at 40 bpm with metronome; (2) 3 reps	%MVIC Time broadness	%MVIC: Pull: -MDI < C PM, MD, BB Forward punch: -MDI < C PM, AD, MD, BB, TB -MDI > C PD, UT Elevation: -MDI < C AD, BB, TB -MDI > C IS Slow overhead throw: -MDI > C PD, UT, IS Rapid overhead throw: -MDI < C BB -MDI > C UT, IS Time broadness: -slow overhead throw MDI > C -rapid overhead throw MDI > C
Illyés et al. (2009) [30]	Nonrandomized controlled trial	P1: 34 MDI treated conservatively; 16 ♀ 18 ♂ P2: 31 MDI treated operatively; 19 ♀ 12 ♂ C: 25 normal, healthy shoulders; 9 ♀ 16 ♂	ZEBRIS surface EMG: monopolar surface electrodes on: -PM -IS -AD, MD, PD -UT -BB -TB	(1) 3 isokinetic movements: -pull in sagittal plane -forward punch in sagittal plane -elevation in scapular plane (2) 3 dynamic movements: -slow overhead throw -rapid overhead throw at maximal speed Modalities: (1) 3 reps at 40 bpm with metronome; (2) 3 reps	%MVIC Time broadness	%MVIC: Pull: -MDI* < C PM, MD, BB Forward punch: -MDI* < C PM, AD, MD, BB, TB -MDI* > C PD, UT Elevation: -MDI* < C AD, BB -MDI* > C IS Slow overhead throw: -MDI* > C PD, UT, IS Rapid overhead throw: -MDI* > C BB, UT, IS Time broadness: -slow overhead throw MDI* > C -rapid overhead throw MDI* > C

Table 2 (*Continued*)

Author (year)	Study design	Study population	Research protocol		Main outcome measures	Main results
			Measurements	Task		
Illyés and Kiss (2006) [38]	Case control	P: 15 MDI; 5 ♀ 10 ♂ C: 15 normal, healthy shoulders; 5 ♀ 10 ♂	ZEBRIS computer-controlled ultrasound-based motion analysis system with EMG: mono-polar surface electrodes on: -PM -IS -AD, MD, PD -SS with trapezius -BB -TB -LT -SA -LD -SCM	Abduction in scapular plane 0°–100/120° with elbow maximally extended and lower arm pronated during entire movement	Kinematics: -GH angle -ST angle -HE angle -GH ratio -ST ratio -SG ratio -ST rhythm -GH rhythm -dSHmin, max -DSH -εSH, εSHx, εShy, εSHz EMG: -muscle actvity length	Kinematics: HE angle: -MDI < C final position, range of angle -MDI > C initial position ST angle: -MDI < C initial and final position, range of angle: -ST ratio, SG ratio: MDI > C -dSHmin, dSHmax: MDI < C -DSH: MDI > C -εSH: MDI > C -εSHx, εSHz: MDI > C EMG: -shorter activity: PM, AD, MD, PD -longer activity: SS, BB, TB, IS
Kiss et al. (2010) [31]	Nonrandomized controlled trial	P1: 32 MDI treated conservatively; 15 ♀ 17 ♂ P2: 19 MDI treated operatively; 12 ♀ 7 ♂ C: 25 controls with healthy shoulders; 9 ♀ 16 ♂	Zebris movement analysis system with EMG: 2 mono-polar surface electrodes on: -AD, MD, PD -SS -trapezius -IS -BB -TB	Elevation in scapular plane 0°–80° with elbow maximally extended and lower arm pronated during entire movement Modalities: 6 reps of elevation per minute, metronome 12 bpm	Kinematics: -GH -ST -HE angle -ST rhythm -GH rhythm -εSH, εSHx, εShy, εSHz EMG: -muscle actvity length	Kinematics: -range of ST: MDI* < C -range of GH: MDI* > C -ST rhythm: slope of regression line MDI* < C -GH rhythm: slope of regression line MDI* > C -εSH: MDI* > C -εSHx: MDI* > C -εSHz: MDI* > C EMG: -MDI* > C: TB, BB, IS, SS -MDI* < C: AD, MD, PD
Morris et al. (2004) [40]	Case control	P1: 5 Multidirectional shoulder laxity P2: 6 MDI shoulders C: 7 normal shoulders	Isokinetic muscle dynamometer and a dual fine-wire (50 μm) EMG into: -SS -IS -AD, MD, PD -subscapularis	5 types of exercise: -ER/IR in neutral -ER/IR in 45° of abduction -ER/IR in 90° of abduction -flexion/extension -Abd/add Modalities: each at 90°/sec, followed by 180°/sec, with 3 submaximal warm-up cycles followed by 3 maximal cycles at each velocity; 2 minutes rest between exercises	Patterns of EMG activity -time course: -individual time points Mean torque of each stage Mean torque ratio	Individual time points: -MD: ER in 90°abd at 90°/s MDI < C -PD: ER/IR in 90°abd at 180°/sec IR half of movement MDI < C -AD: ER/IR in neutral at 180°/sec MDI > C; ER in 90° abd at 90°/sec MDI < C Time course: -MD: ER/IR in 90°abd at 90°/sec, ER half of movement MDI < C -PD: ER/IR in 90°abd at 180°/sec MDI = C -AD: ER/IR in 90° at 90°/sec MDI < C for ER; MDI = C for overall & IR Torque data: IR/ER ratio, add/abd ratio MDI > C

Table 2 (*Continued*)

Author (year)	Study design	Study population	Research protocol		Main outcome measures	Main results
			Measurements	Task		
Nyiri et al. (2010) [32]	Nonrandomized controlled trial	P1: 32 MDI treated conservatively; 12 ♀ 15 ♂ P2: 19 MDI treated operatively; 9 ♀ 5 ♂ C: 25 controls with healthy shoulders; 9 ♀ 16 ♂	Zebris movement analysis system with EMG; surface electrodes on: -AD, MD, PD -SS -IS -BB -TB	Elevation in scapular plane 0°–80° with elbow maximally extended and lower arm pronated during entire movement Modalities: 6 reps of elevation per minute, metronome 12 bpm	Kinematics: -GH angle -ST angle -HE angle -ST rhythm -GH rhythm -εSH, εSHx, εShy, εSHz EMG: -muscle activity length	Kinematics: -range of ST: MDI* < C -range of GH: MDI* > C -GH rhythm: slope of regression line MDI* > C -ST rhythm: slope of regression line MDI* < C -εSH, εSHx, εSHz: MDI* > C EMG: -MDI* > C: TB, BB, IS, SS -MDI* < C: AD, MD, PD
Ogston and Ludewig (2007) [34]	Case control	P: 29 MDI; 13 ♀ 16 ♂ C: 29 controls with healthy shoulders; 13 ♀ 16 ♂	Polhemus FASTRAK electromagnetic motion capture system. Lightweight sensors applied to: -thorax (manubrium sterni) -scapula (acromion process) -humerus (distal)	Elevation in frontal and scapular plane through their perceived full range of motion using 6-seconds verbal count (3sec up, 3sec down) Modalities: 5 consecutive reps; 30sec rest between each motion	Scapular IR Scapular upward rotation Scapular tipping Humeral anterior-posterior Humeral superior inferior translation	Elevation in scapular plane: -scapular IR MDI > C -scapular upward rotation MDI < C -scapular posterior tipping MDI > C -humeral translations MDI = C Elevation in frontal plane: -scapular upward rotation MDI < C -humeral translations MDI = C
Sciascia et al. (2012) [39]	Case control	P1: 10 generalized laxity (GL) P2: 10 MDI P3: 9 anterior instability C: 10	2 bipolar surface electrodes on: -UT -MD -SA 2 sterile bipolar fine-wire electrodes were inserted into: -SS -IS -TM Photocell sensor system synchronized to EMG data to determine specific arcs of motion	(1) Open kinetic chain exercises: -120° scaption with ER; mean load 2 (0.7) kg -90° prone horizontal abd; mean load 1.5 (0.5) kg -Prone ER at 90° abduction; mean load 1.9 (0.8) kg (2) Closed kinetic chain: -push-up plus Modalities: (1) 6 reps at 50°/sec with metronome; (2) 6 reps at subjects' self-selected pace (1rep/2sec)	%MVIC	Scaption: -TM (eccentric 60–30° arc): GL > C = MDI (very low amplitudes, probably not clinically relevant) 90° prone horizontal abd: -IS (concentric 30–60° arc): GL > MDI -TM (concentric 30–60° arc): MDI < C Prone ER: -IS MDI > C tendency ($P = 0.07$) Push-up plus: MDI = C
Von Eisenhart-Rothe et al. (2005) [42]	Case control	P: 14 atraumatic shoulder instability with multidirectional laxity; 9 ♀ 5 ♂ C: 28 healthy shoulders; 16 ♀ 12 ♂	Open MRI system and a T1-weighted 3D gradient recalled echo sequence with image acquisition in transversal orientation perpendicular to the cavitas glenoidalis	-humerus passively positioned relative to the thorax at: -30° abd in scapular plane -90° abd -90° abd ER -maximal add Isometric (10 N) muscle activity: -90° abd with external rotation (ISOM)	Mean SH rhythm ratio (SHR) Scapular internal rotation (degrees) 3D position (mm) of the humeral head relative to the glenoid cavity Coefficient of correlation between scapular positioning and humeral head position	SHR: -MDI > C in 30° abd and 90° abd -MDI = C in 90° abd IR and ISOM Scapular internal rotation: MDI > C in 30° abd and ISOM 3D position: -ISOM: C = recentering of humeral head ↔ MDI = decentered humeral head position in both planes -MDI had malcentering of the humeral head in the direction of instability during specific arm positions Correlation: In all shoulders (affected, unaffected and healthy) a high correlation was found during muscle relaxation in both planes, especially at 30° and 90° abd

Table 2 (*Continued*)

Author (year)	Study design	Study population	Research protocol		Main outcome measures	Main results
			Measurements	Task		
Von Eisenhart-Rothe et al. (2010) [41]	Case control	P: 14 atraumatic shoulder instability with multidirectional laxity; 9 ♀ 5 ♂ C: 28 healthy shoulders; 16 ♀ 12 ♂	Open MRI system and a T1-weighted 3D gradient recalled echo with image acquisition in transversal orientation perpendicular to the cavitas glenoidalis	Humerus passively positioned relative to the thorax at: -30° abd in scapular plane -90° abd -90° abd ER Maximal add Isometric (10 N) muscle activity: -90° abd with external rotation (ISOM)	Glenoid retroversion Glenoid size HHR 3D GH index Curvature analysis Glenoid radius SHR Scapular internal rotation 3D position of humeral head relative to the glenoid cavity	Retroversion: MDI > C Glenoid size, HHR, 3D GH index: MDI < C tendency ($P = 0.17$) Curvature: MDI < C Glenoid radius: MDI > C SHR: MDI > C in 30° abd and 90° abd Scapular internal rotation: MDI > C in 30° abd and ISOM 3D position: -ISOM: C = recentering of humeral head ↔ MDI = decentered humeral head position in both planes -MDI had malcentering of the humeral head in the direction of instability during specific arm positions
Ozaki (1989) [36]	Case control	P: 11 involuntary inferior and multidirectional instability; 4 ♀ 7 ♂ C: 30 normal shoulders; 15 ♀ 15 ♂	Cineradiography recorded in 8-mm cinefilms	Abd in the scapular plane with the arm in neutral rotation	SCE angle Glenoid angle	-SCE angle MDI < C -glenoid angle MDI > C
Santos et al. (2007) [33]	Case control	P: 8 swimmers with history of shoulder impingement and at least one kind of GH instability without pain; 4 ♀ 4 ♂ C: 8 swimmers with normal shoulders; 4 ♀ 4 ♂	-surface electrodes on AD PM superior portion LD SA UT LT BB TB long head -optotrak LEDs markers fixed on SIAS Ant. aspect humeral head Ant. articular line elbow joint	Bilateral and simultaneous: -elevation of both shoulders in -scapular plane with elbow -extended and forearm in neutral position Modalities: movements were executed into 3 different target distances: 30°, 90°, 150° Movements were performed 5 × toward each target as fast as possible	Kinematics: -shoulder angle -shoulder velocity -shoulder translational movement: x,y,z EMG: -muscle activity onset	Kinematics: MDI = C EMG: MDI = C

* Showing results for MDI patients before treatment. %MVIC: percentage muscle activity normalized by maximal voluntary isometric contraction; abd, abduction; ACT: duration of activation in terms of range of motion; AD: anterior deltoid; add, adduction; AMP: normalized average amplitude; Ant.: anterior; BB: biceps brachii; bpm, beats per minute; C: controls; DSH: absolute displacement between rotation center of scapula and humerus; dSHmax: longest distance (mm) between rotation centers of scapula and humerus; dSHmin: smallest distance (mm) between rotation centers of scapula and humerus; EMG: electromyography; ER: external rotation; εSH: relative displacement between rotation center of scapula and humerus; εSHx: εShy, εSHz, components in directions x, y, and z of the relative displacement between rotation center of scapula and humerus; GH: glenohumeral; GL: generalized laxity; HE: humeral elevation; HHR: humeral head radius; IR: internal rotation; IS: infraspinatus; ISOM: isometric (10 N) muscle activity at 90° abduction with external rotation; LEDs: light-emitting diodes; LD: latissimus dorsi; LT: lower trapezius; MD: middle deltoid; MDI: multidirectional shoulder instability; ONST: onset of activation; P: patients; PD: posterior deltoid; PM: pectoralis major; reps, repetitions; SA: serratus anterior; SCE: shoulder center edge; SCM: sternocleidomastoideus; sec: seconds; SG: scapuloglenoid; SHR: mean SH rhythm ratio; SIAS: anterior superior iliac spine; SS: supraspinatus; ST: scapulothoracic; TERM: termination of activation; TM: teres major; TB: triceps brachii; UT: upper trapezius.

data were extracted, and the case–control checklist was used to assess the risk of bias. Two researchers independently scored each article. When an item was fulfilled, articles earned one point (+). Zero points were given if an item was not met (−) or if the article did not provide sufficient information to answer the question (?). The scores of both researchers were compared and a final score was assigned during a consensus meeting. Each article was graded with a level of evidence according to the CBO guidelines. The data extracted from included articles was summarized in a qualitative manner.

3. Results

3.1. Study selection

The initial search identified 403 potentially relevant articles. After screening on the title and abstract, 23 articles remained for further evaluation. After screening on the full text, 12 eligible articles remained. One additional study was obtained by hand search [33], thus resulting in 13 included articles for analysis. The selection process is illustrated in Fig. 1.

3.2. Risk of bias and level of evidence

The risk of bias and level of evidence are summarized in Table 3. The authors agreed in 96% of cases. A level of evidence B was attributed to all studies. Case-control studies received quality scores ranging from 5/6 (83%, low risk of bias) to 0/6 (0%, high risk of bias). Two articles were high quality [34,35], 8 moderate, and 3 poor [30,36,37]. All studies lost points on item 5 because they did not describe whether the researchers were blinded to participant group. The articles of Nyiri et al. [32] and Kiss et al. [31] seemed based on an identical group of MDI individuals and yielded identical results and were therefore considered a single study.

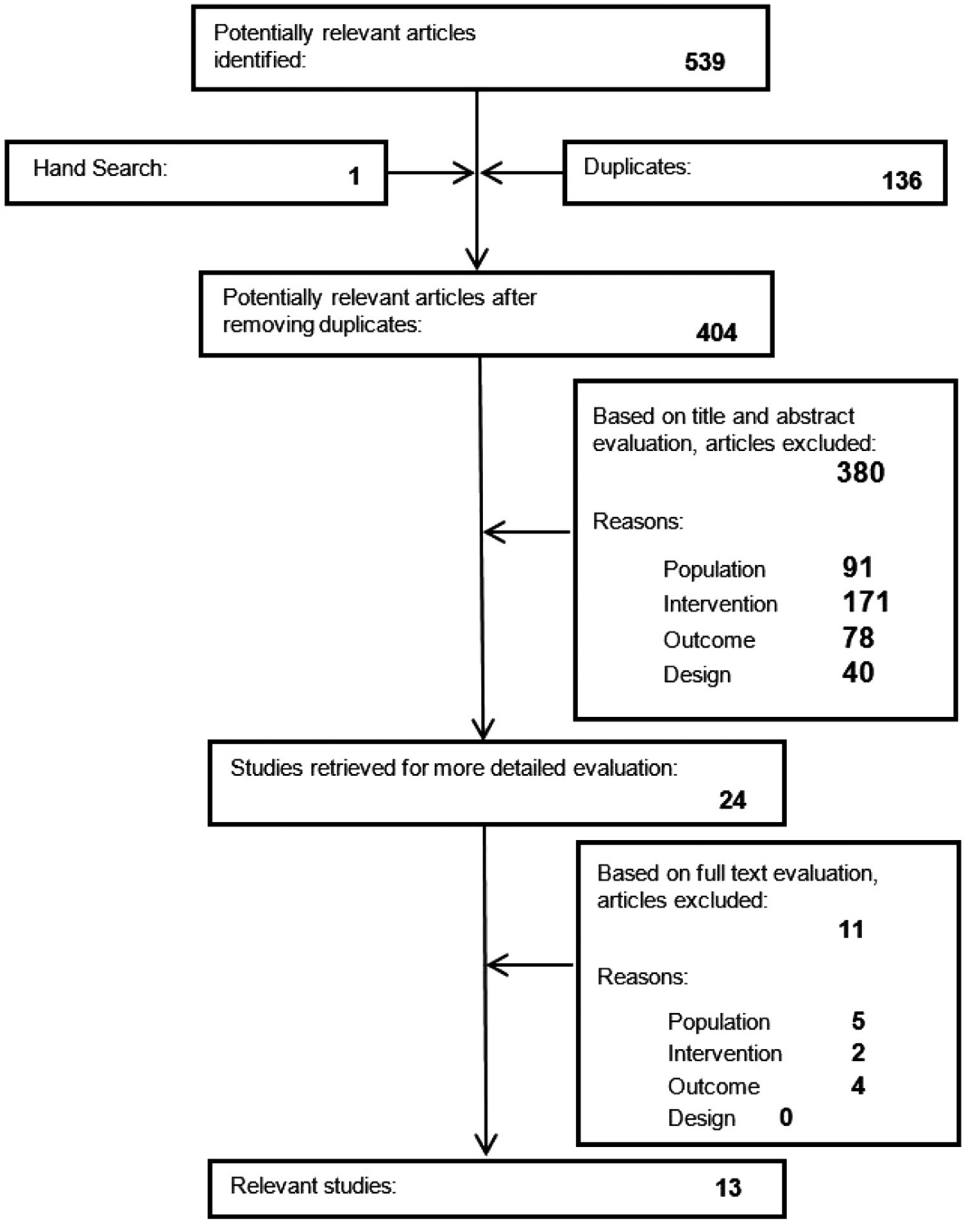


Fig. 1. Flowchart of the study selection.

Table 3
The risk of bias and level of evidence of included studies.

Author (year)	1	2	3	4	5	6	Score	Score (%)	Level of evidence
Barden et al. (2005)	+	+	+	+	?	+	5/6	83	B
Illyés and Kiss (2007)	–	–	?	+	?	–	1/6	17	B
Illyés and Kiss (2006)	+	+	+	+	?	–	4/6	67	B
Morris et al. (2004)	+	–	+	+	?	–	3/6	50	B
Ogston and Ludewig (2007)	+	+	+	+	?	+	5/6	83	B
Sciascia et al. (2012)	+	+	+	+	?	–	4/6	67	B
Von Eisenhart–Rothe et al. (2005)	+	+	+	+	?	–	4/6	67	B
Von Eisenhart–Rothe et al. (2010)	+	+	+	+	?	–	4/6	67	B
Santos et al. (2007)	+	+	–	+	?	–	3/6	50	B
Illyés et al. (2009)	–	+	–	+	?	–	2/6	33	B
Kiss et al. (2010)	+	+	+	+	–	–	4/6	67	B
Nyiri et al. (2010)	+	+	–	+	?	–	3/6	50	B
Ozaki et al. (1987)	–	–	–	–	–	–	0/6	0	B

– Score not fulfilled, + score fulfilled; ? answer is unclear. Higher scores imply a higher methodological quality and lower risk of bias. 1: are the patients clearly and adequately defined? 2: are the controls clearly and adequately defined? 3: can selection bias be sufficiently excluded? 4: is procedure exposure clearly defined and is the method of assessment to exposure adequate? 5: is the exposure blindly recorded for pathology? 6: are the most important confounders identified and are they taken into account in the investigation or in the analysis?

3.3. Study characteristics

Study characteristics are displayed in the evidence table (Table 2). Nine of 13 articles examined arm elevation in the scapular plane with range of motion from 80° to 150° [30–34,36–39]. In these studies, the definition of the scapular plane varied from 20° to 45° anterior to the frontal plane. Furthermore, 3 articles studied abduction in the frontal plane [34,35,40], and other studies assessed glenohumeral internal or external rotation [35,39,40]. Flexion/extension movements in the sagittal plane were evaluated in 2 studies [35,40]. More functional movements such as pushing, pulling, slow and rapid overhead throws were investigated in 3 publications [30,37,39]. Finally, 2 articles examined the position of the scapula and humeral head during a series of passive arm positions and during isometric contraction [41,42]. For the passive tests, participants were positioned supine on a pad that allowed free movement of the scapula, with the humerus in various degrees of abduction, supported by a board. In the isometric condition, the arm was supported in 90° of abduction with a weight attached to the lower arm to provoke an external rotation force of 10 N, while the person was asked to hold the arm in place by working against this force. The goal was to assess the effect of isometric muscle activity on humeral head translations [41,42].

To gain insight into the muscle activation pattern in MDI individuals, 9 articles used surface electromyography (EMG) for several muscles: the deltoideus [30–33,35,37–40], latissimus dorsi (LD) [33,35,38], pectoralis major (PM) [30,33,35,37,38], infraspinatus [30–32,37,38], supraspinatus [31,32,38], biceps brachii (BB) [30–33,37,38], triceps brachii (TB) [30–33,37,38], serratus anterior (SA) [38,39], upper (UT) [30,33,37,39] or lower trapezius (LT) [33,38], and sternocleidomastoideus [38]. Fine wire EMG was performed in 3 studies to assess the activity of the infraspinatus [35,39,40] and supraspinatus [35,39,40], deltoideus [40], subscapularis [40] and teres major [39]. Four of 10 articles [30,35,37,39] reported on muscle activity (generally expressed as a percentage of the activity during maximal voluntary isometric contraction [%MVIC]), and 7 reported parameters that reflect timing of activity (e.g., muscle activity length, time broadness, onset of muscle activity) [30–33,35,37,38].

To track the movement pattern of the humerus and scapula (kinematic data), 3 studies used ZEBRIS (ultrasound-based tracking) [31,32,38], one FASTRAK (electromagnetic tracking) [34], and 2 an open MRI system [41,42]. An older study by Ozaki et al. (1989) used cineradiograms to assess the position of the scapula and humeral head during arm elevation [36], a technique in which sequential radiographs are made of moving body structures, so that the movement can be projected as a motion picture [43].

Three studies evaluated scapular upward rotation by using the scapulothoracic (ST) angle (Table 4) [31,32,38], 2 by reporting the scapulohumeral (SH) ratio or rhythm (Table 4) [41,42], one by reporting the shoulder center edge angle [36], and one by calculating the angle between the local coordinate system of the scapula and thorax, as defined by the International Society of Biomechanics recommendations [34]. Three studies reported scapular internal rotation [34,41,42] and one anteroposterior tilt [34]. Scapular rotations are depicted in Fig. 2. To describe humeral elevation, most studies used the humeral elevation (HE) angle, and/or the glenohumeral (GH) angle (Table 4).

In addition, several ratios between these angles were reported, such as the GH rhythm [31,32,38], ST rhythm [31,32,38] and SH rhythm [41,42]. Six articles attempted to describe the centering and translatory movements of the humeral head, for instance by reporting the displacement between the rotation center of the humerus and the rotation center of the scapula [31,32,38], by measuring the anteroposterior and craniocaudal movement of the humeral head relative to the glenoid [41,42] or by evaluating the shoulder center edge angle [36]. All reported kinematic outcome parameters are explained in Table 4.

4. Synthesis of results

4.1. EMG results

4.1.1. Muscle activity

4.1.1.1. Arm elevation in the scapular plane

In 3 articles, EMG activity during elevation in the scapular plane was reported as %MVIC [30,37,39]. An overview of the investigated muscles and results can be found in the evidence table (Table 2). Two studies found significantly reduced activity in the BB [30,37] and anterior deltoid (AD) [30,37] in people with MDI. Furthermore, 2 studies reported significantly increased activity in the infraspinatus [30,37], but in one study, the difference with controls did not reach statistical significance [39]. No significant differences were found between MDI individuals and controls for the other rotator cuff muscles and other scapular muscles.

4.1.1.2. Pull and forward punch

Two studies found reduced EMG activity during a pull and forward punch in the BB, middle deltoid (MD), and PM

Table 4
Summary of kinematic outcome parameters of the humerus and scapula.

Angle between humerus and thorax	Humeral elevation angle (HE)	Illyés & Kiss 2006, Kiss et al., 2010, Nyiri et al., 2010: Angle between: The thorax: a line between the proximal and distal point of the sternum The humerus: a line between the insertion point of the deltoideus and the radial epicondyle
	Arm abduction angle	Von Eisenhart-Rothe et al., 2005, Von Eisenhart-Rothe et al., 2010: Angle between: The thorax: spine The humerus: longitudinal axis of the humerus
Angle between humerus and scapula	Glenohumeral angle (GH)	Illyés & Kiss 2006, Kiss et al., 2010, Nyiri et al., 2010 Humerus: a line between the insertion point of the deltoideus and the radial epicondyle Scapula: a line between the angulus acromialis and the trigonum spinae
	Glenohumeral angle (GH)	Von Eisenhart-Rothe et al., 2005, Von Eisenhart-Rothe et al., 2010: Angle between: Humerus: longitudinal axis Scapula: glenoid surface
Angles reflecting upward rotation of the scapula	Scapulothoracic angle (ST)	Illyés & Kiss 2006, Kiss et al., 2010, Nyiri et al., 2010 Angle between: Thorax: a line between the proximal and distal point of the sternum Scapula: a line between the angulus acromialis and the trigonum spinae
	Scapulothoracic angle (ST)	Von Eisenhart-Rothe et al., 2005, Von Eisenhart-Rothe et al., 2010: Angle between: Thorax: spinal axis Scapula: glenoid plane
	Scapular upward rotation (UR)	Ogston & Ludewig 2007 Rotation of the scapula around its local Y-axis, as described in the ISB recommendations
	Glenoid angle	Ozaki 1989 Angle between the glenoid surface (line between the most cranial and caudal point) and a horizontal line.
Glenohumeral ratio or rhythm (GH rhythm)	Ratio between the glenohumeral angle and the humeral elevation angle	Illyés & Kiss 2006, Kiss et al., 2010, Nyiri et al., 2010 GH/HE The movement amplitude between the humerus and scapula, per degree of arm elevation
Scapulothoracic ratio or rhythm (ST rhythm)	Ratio between the scapulothoracic angle and the humeral elevation angle	Illyés & Kiss 2006, Kiss et al., 2010, Nyiri et al., 2010 ST/HE The movement amplitude between the scapula and thorax, per degree of arm elevation
Scapulohumeral ratio or rhythm (SH rhythm)	Ratio between the glenohumeral and scapulothoracic angle	Von Eisenhart-Rothe et al., 2005, Von Eisenhart-Rothe et al., 2010: GH/ST Ratio between humeral and scapular movement

[30,37]. Additionally, during forward punching, MDI individuals showed reduced EMG activity in the AD and TB [30,37], but the muscle activity in the posterior deltoid (PD) and UT was increased [30,37]. Finally, infraspinatus activity did not significantly differ between MDI individuals and controls, nor during pulling nor forward punching [30,37].

4.1.1.3. Slow and rapid overhead throw

We found moderate evidence that the infraspinatus and UT are more active in MDI individuals than controls during slow and rapid overhead throws [30,37]. Additionally, during slow overhead throwing, MDI individuals activated the PD more than did controls [30,37]. Regarding biceps activity, the 2 studies were contradictory [30,37].

4.1.2. Timing of muscle activity

Two studies compared the onset of muscle activation [33,35]. Santos et al. [33] found that the time of onset of the AD, PM, LD, SA, UT, LT, BB, and long head of the triceps did not significantly differ between swimmers with MDI and healthy swimmers during bilateral elevation in the scapular plane (Table 2). However, Barden et al. [35] found that the onset of activation of the

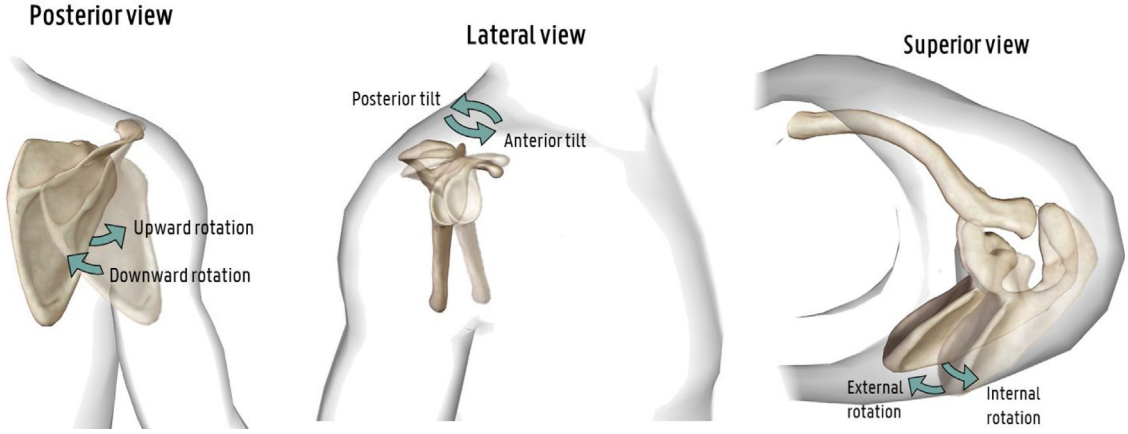


Fig. 2. Scapular kinematics (3-D rotations).

PD was significantly delayed in MDI individuals during external rotation.

Three studies evaluated the muscle activity length during elevation in the scapular plane [31,32,38]. All 3 reported a significantly longer activity of infraspinatus, supraspinatus, BB and TB in MDI versus controls. Furthermore, MDI individuals showed shorter activity of the PM [38] and all 3 parts of the deltoid [31,32].

Two articles studied time broadness, which reflects the time elapsed between the peak activity of the muscle that was active first and the muscle that was active last during the movement cycle [30,37]. Both authors reported that time broadness was significantly longer in the MDI group during both slow and rapid overhead throws, which indicates that the muscles of individuals with MDI are active during a larger part of the movement cycle [30,37].

Only one study reported the termination of muscle activation and the duration of muscle activation relative to the range of motion [35]. Barden et al. found a shortened supraspinatus and infraspinatus activation during external rotation in MDI individuals [35]. In addition, during shoulder extension, the termination of PM activity occurred later in MDI individuals than controls. Moreover, MDI individuals activated the PM throughout a greater part of range of motion [35].

4.2. Kinematic results

4.2.1. Scapular movement pattern

Multiple studies found decreased upward rotation of the scapula in MDI individuals during elevation of the arm in the scapular plane (decreased ST angle and "ST angle range") [31,32,34,38,41,42]. Furthermore, Ogston and Ludewig [34] and Von Eisenhart-Rothe et al. [43,44] found that MDI individuals showed a larger scapular internal rotation compared to controls, during arm elevation in the scapular plane and when the arm was passively placed in several positions. Finally, Ogston and Ludewig also found more posterior scapular tilting in MDI than controls during the first 2 phases (0°–30° and 31°–60°) of arm elevation [34].

4.2.2. Angular movement of the humerus, relative to the scapula

Both Kiss et al. and Nyiri et al. reported that the GH angle (Table 4) was significantly larger in MDI individuals than controls during arm elevation in the scapular plane [31,32], which suggests that the MDI group showed increased humeral movement relative to the scapula. Three studies also reported GH and ST rhythms during elevation in the scapular plane (Table 4) [31,32,38]. All 3 articles agreed that GH and ST rhythms of MDI individuals differed from controls. The graphical relationship between the scapular upward rotation and the angle of arm elevation was more linear in MDI individuals than controls but was rather bilinear in controls with an angulation around 60° of arm elevation. Two studies assessed the SH rhythm (Table 4) [41,42]. The affected shoulders of MDI individuals showed an increased SH rhythm when the arm was passively positioned in 30° of abduction in the scapular plane and 90° of abduction in the frontal plane [41,42]. This finding suggests more humeral elevation and/or less scapular upward rotation.

4.2.3. Translatory movement of the humeral head, relative to the scapula

In 3 studies, the movement of the scapula and the humerus relative to each other was described by the relative displacement between the rotational centers of the humerus and scapula during arm elevation [31,32,38]. MDI shoulders showed increased mean relative displacement between rotation centers as compared with healthy shoulders [31,32,38] as well as between the affected and unaffected MDI shoulder [38]. Accordingly, Von Eisenhart-Rothe et al. found a significant decentering of the humeral head in MDI individuals by using MRI, both in passive abduction (at 30° and 90°) and during isometric muscle activation [41,42]. Also Ozaki et al. found that the humeral head had an excessive excursion in MDI individuals [36]. In contrast to these studies, Ogston and Ludewig and Santos et al. could not detect altered humeral head translation in MDI [33,34].

5. Discussion

This systematic literature review summarizes the evidence of 13 original research articles regarding muscle activity and shoulder kinematics in individuals with MDI versus healthy controls. Alterations in muscle activity, timing of muscle activation, scapular movement patterns, and humeral head translation were found.

5.1. EMG results

Taken together, the EMG results suggest that individuals with MDI may have [1] an increased or prolonged activity of muscles that stabilize the humeral head; [2] a reduced or shorter activity of muscles that move or accelerate the arm and shoulder girdle; [3] and increased or prolonged activity of muscles that decelerate or eccentrically control the movement of the arm and shoulder girdle. Which muscles accelerate or decelerate the arm depends on the specific arm movement under evaluation.

To illustrate the first part of the hypothesis [1], several studies reported increased [30,37] or prolonged [31,32,38] infraspinatus activity and prolonged supraspinatus activity [31,32,38] in MDI individuals during elevation in the scapular plane. Because these rotator cuff muscles create compressive forces that prevent excessive glenohumeral translations [23,44,45], their increased or prolonged activity presumably reflects an attempt to control the humeral head better and longer [46]. By contrast, 3 other studies could not confirm the increased/prolonged involvement of the supraspinatus or infraspinatus [35,39,40]. In these studies, the sample sizes may have been too small (respectively 7, 10, and 6 participants) [35,39,40], and in Sciascia et al. [39] and Morris et al. [40], the number of included groups too large to detect differences between the MDI and control group for these muscles (4 and 3 groups, respectively).

Besides glenohumeral muscles, there is currently insufficient evidence to state that the activity of scapular stabilizers is prolonged or increased as well. The activity of the LT, SA, and LD was evaluated in several studies but did not significantly differ between MDI individuals and controls [33,35,38,39], although they are thought to play a key role in scapular stabilization [26].

Regarding the second hypothesis [2], there is moderate evidence to suggest that patients with MDI demonstrate a reduced (or sometimes shortened) activity of muscles that move or accelerate the arm. For instance, during elevation in the scapular plane, 2 studies demonstrated decreased muscle activity of the BB and decreased and shortened activation of the AD [30,37], both prime movers during this motion. Likewise, during forward punching, the prime movers (AD, MD, BB and TB) showed reduced activity in MDI individuals [30,37].

A third hypothesis [3] emerging from this review is that muscles that decelerate the movement of the arm and shoulder girdle appear to be engaged more and/or longer in MDI individuals than controls. As an illustration, the increased activity of the PD and UT during forward punching can be interpreted as an attempt to control the arm and scapular movement actively in an eccentric manner, to slow down the forward movement of the arm and shoulder girdle and prevent anterior (sub)luxation of the humeral head [37].

5.2. Kinematic results

In patients with MDI, the research strongly suggests insufficient upward scapular rotation during arm elevation [31,32,34,36,38]. One of the questions that arise is whether more downward rotation also facilitates a larger inferior translation of the humeral head, which is also prevalent in MDI [2,5,19]. Less scapular upward rotation theoretically allows the humeral head to slide inferiorly more easily in case of ligamentous laxity [47,48]. As such, it has been mentioned as a factor that may contribute to the mechanism of glenohumeral (sub)luxation [7,34,49,50]. In rehabilitation, shrug exercises have been recommended, to stimulate UT activity [6,10] to increase scapular upward rotation [51].

Finally, MDI individuals might have increased scapular internal rotation during arm elevation in the scapular plane. This is in line with the clinical sign of scapular "winging", defined as a prominence of the medial border of the scapula [52], often observed in MDI [2,53,54]. Increased internal rotation and decreased upward rotation in MDI suggest deficiencies in neuromuscular control [34], which could be addressed in treatment [6,14,54].

5.3. Strengths and limitations

We identified several strengths and limitations in the current literature. First, the results on muscle activity (%MVIC) should be interpreted carefully because several of the statistically significant differences between patients and controls emerged from articles with a high risk of bias [30,37]. Future studies of high methodological quality are needed to confirm these EMG findings. Another reason for caution in interpreting EMG measurements is that the standard normalization procedure to a %MVIC may not be suited for MDI individuals because maximal muscle contractions often cause discomfort [55,56]. Testing should occur in a pain-free or reduced-pain condition to avoid underestimation of the MVIC and consequent overestimation of the muscle activity during arm movements [55]. Hence, future studies evaluating the validity of normalization to a %MVIC and the reliability of alternative submaximal normalization methods are needed to draw valid conclusions in patients with MDI.

The most consistent finding in this review is the alteration in scapular kinematics in MDI, reported by articles of moderate to high quality. However, despite the agreement on scapular kinematics, surprisingly, most studies could not detect any significant group differences in muscle activity between people with MDI and controls. One reason may be that the MDI population is heterogenous and that there are several different aberrant muscle patterns that can lead to the same pattern of insufficient scapular upward rotation. Future studies should include larger sample sizes, to identify subgroups with different muscle patterns in MDI patients.

A third limitation is that the reviewed studies mostly compared data points and averages of specific movement amplitudes between individuals with MDI and controls. More recent statistical techniques, such as statistical parametric mapping, now allow for comparing EMG and kinematic data over the full range of motion [57].

Finally, despite the predefined eligibility criteria, included articles still differed regarding MDI inclusion criteria, as well as methodology. Therefore, we strongly recommend that future studies use a more detailed description of participants and more similar outcome measures.

6. Conclusions

The reviewed literature strongly suggests that individuals with MDI show less upward rotation and more internal rotation of the scapula than controls. Furthermore, some evidence suggests increased or prolonged activation of muscles that stabilize the humeral head, decreased or shorter activation of muscles that move or accelerate the arm and shoulder girdle, and increased or prolonged activation of muscles that decelerate the movement of the arm and shoulder girdle. These hypotheses constitute starting points for future research and may provide indications for exercise therapy.

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Disclosure of interest

The authors declare that they have no competing interest.

References

[1] Abboud JA, Soslowsky LJ. Interplay of the static and dynamic restraints in glenohumeral instability. Clin Orthop Relat Res 2002;400:48–57.
[2] Guerrero P, Busconi B, Deangelis N, Powers G. Congenital instability of the shoulder joint: assessment and treatment options. J Orthop Sports Phys Ther 2009;39:124–34.
[3] Kuhn JE. A new classification system for shoulder instability. Br J Sports Med 2010;44:341–6.
[4] Ren H, Bicknell RT. From the unstable painful shoulder to multidirectional instability in the young athlete. Clin Sports Med 2013;32:815–23.
[5] An YH, Friedman RJ. Multidirectional instability of the glenohumeral joint. Orthop Clin North Am 2000;31:275–85.
[6] Cools AM, Borms D, Castelein B, Vanderstukken F, Johansson FR. Evidence-based rehabilitation of athletes with glenohumeral instability. Knee Surg Sports Traumatol Arthrosc 2016;24:382–9.
[7] Neer CS, Foster CR. Inferior capsular shift for involuntary inferior and multidirectional instability of the shoulder. A preliminary report. J Bone Joint Surg Am 1980;62:897–908.
[8] Bahu MJ, Trentacosta N, Vorys GC, Covey AS, Ahmad CS. Multidirectional instability: evaluation and treatment options. Clin Sports Med 2008;27:671–89.
[9] Johansson K. Multidirectional instability of the glenohumeral joint: an unstable classification resulting in uncertain evidence-based practice. Br J Sports Med 2016;50:1105–6.
[10] Warby SA, Ford JJ, Hahne AJ, Watson L, Balster S, Lenssen R, et al. Comparison of 2 exercise rehabilitation programs for multidirectional instability of the glenohumeral joint: a randomized controlled trial. Am J Sports Med 2018;46:87–97.
[11] Richards RR. The diagnostic definition of multidirectional instability of the shoulder: searching for direction. J Bone Joint Surg Am 2003;85-a:2145–6.
[12] Mallon WJ, Speer KP. Multidirectional instability: current concepts. J Shoulder Elbow Surg 1995;4:54–64.
[13] McFarland EG, Kim TK, Park HB, Neira CA, Gutierrez MI. The effect of variation in definition on the diagnosis of multidirectional instability of the shoulder. J Bone Joint Surg Am 2003;85-a:2138–44.
[14] Ruiz Iban MA, Diaz Heredia J, Garcia Navlet M, Serrano F, Santos Oliete M. Multidirectional Shoulder Instability: Treatment. Open Orthop J 2017;11:812–25.
[15] Hill A, Bull A, Richardson J, McGregor A, Smith C, Barrett C, et al. The clinical assessment and classification of shoulder instability. Curr Orthop 2008;22:208–25.
[16] Lee SB, Kim KJ, O'Driscoll SW, Morrey BF, An KN. Dynamic glenohumeral stability provided by the rotator cuff muscles in the mid-range and end-range of motion. A study in cadavera. J Bone Joint Surg Am 2000;82:849–57.
[17] Gaskill TR, Taylor DC, Millett PJ. Management of multidirectional instability of the shoulder. J Am Acad Orthop Surg 2011;19:758–67.
[18] Kibler WB, Sciascia A. The role of the scapula in preventing and treating shoulder instability. Knee Surg Sports Traumatol Arthrosc 2016;24:390–7.
[19] Lee HJ, Kim NR, Moon SG, Ko SM, Park JY. Multidirectional instability of the shoulder: rotator interval dimension and capsular laxity evaluation using MR arthrography. Skeletal Radiol 2013;42:231–8.
[20] Spanhove V, De Wandele I, Kjær BH, Malfait F, Vanderstukken F, Cools A. The effect of five isometric exercises on glenohumeral translations in healthy subjects and patients with the hypermobility type of the ehlers-danlos syndrome (heds) or hypermobility spectrum disorder (hsd) with multidirectional shoulder instability: an observational study. Physiotherapy 2019;107:11–8.
[21] Matsumura N, Oki S, Fukasawa N, Matsumoto M, Nakamura M, Nagura T, et al. Glenohumeral translation during active external rotation with the shoulder abducted in cases with glenohumeral instability: a 4-dimensional computed tomography analysis. J Shoulder Elbow Surg 2019;28:1903–10.
[22] Gaskill TR, Braun S, Millett PJ. The rotator interval: pathology and management. Arthroscopy 2011;27:556–67.
[23] Wattanaprakornkul D, Cathers I, Halaki M, Ginn KA. The rotator cuff muscles have a direction specific recruitment pattern during shoulder flexion and extension exercises. J Sci Med Sport 2011;14:376–82.

[24] Forthomme B, Crielaard JM, Croisier JL. Scapular positioning in athlete's shoulder: particularities, clinical measurements and implications. Sports Med 2008;38:369–86.
[25] Kibler WB, Ludewig PM, McClure PW, Michener LA, Bak K, Sciascia AD. Clinical implications of scapular dyskinesis in shoulder injury: the 2013 consensus statement from the 'Scapular Summit'. Br J Sports Med 2013;47:877–85.
[26] Cools AM, Struyf F, De Mey K, Maenhout A, Castelein B, Cagnie B. Rehabilitation of scapular dyskinesis: from the office worker to the elite overhead athlete. Br J Sports Med 2014;48:692–7.
[27] Weiser WM, Lee TQ, McMaster WC, McMahon PJ. Effects of simulated scapular protraction on anterior glenohumeral stability. Am J Sports Med 1999;27:801–5.
[28] Itoi E. Pathophysiology and treatment of atraumatic instability of the shoulder. J Orthop Sci 2004;9:208–13.
[29] Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. Syst Rev 2015;4:1.
[30] Illyes A, Kiss J, Kiss RM. Electromyographic analysis during pull, forward punch, elevation and overhead throw after conservative treatment or capsular shift at patient with multidirectional shoulder joint instability. J Electromyogr Kinesiol 2009;19:e438–47.
[31] Kiss RM, Illyes A, Kiss J. Physiotherapy vs. capsular shift and physiotherapy in multidirectional shoulder joint instability. J Electromyogr Kinesiol 2010; 20:489–501.
[32] Nyiri P, Illyes A, Kiss R, Kiss J. Intermediate biomechanical analysis of the effect of physiotherapy only compared with capsular shift and physiotherapy in multidirectional shoulder instability. J Shoulder Elbow Surg 2010;19:802–13.
[33] Santos MJ, Belangero WD, Almeida GL. The effect of joint instability on latency and recruitment order of the shoulder muscles. J Electromyogr Kinesiol 2007;17:167–75.
[34] Ogston JB, Ludewig PM. Differences in 3-dimensional shoulder kinematics between persons with multidirectional instability and asymptomatic controls. Am J Sports Med 2007;35:1361–70.
[35] Barden JM, Balyk R, Raso VJ, Moreau M, Bagnall K. Atypical shoulder muscle activation in multidirectional instability. Clin Neurophysiol 2005;116:1846–57.
[36] Ozaki J. Glenohumeral movements of the involuntary inferior and multidirectional instability. Clin Orthop Relat Res 1989;238:107–11.
[37] Illyes A, Kiss RM. Electromyographic analysis in patients with multidirectional shoulder instability during pull, forward punch, elevation and overhead throw. Knee Surg Sports Traumatol Arthrosc 2007;15:624–31.
[38] Illyes A, Kiss RM. Kinematic and muscle activity characteristics of multidirectional shoulder joint instability during elevation. Knee Surg Sports Traumatol Arthrosc 2006;14:673–85.
[39] Sciascia A, Kuschinsky N, Nitz AJ, Mair SD, Uhl TL. Electromyographical comparison of four common shoulder exercises in unstable and stable shoulders. Rehabil Res Pract 2012;2012:783824.
[40] Morris AD, Kemp GJ, Frostick SP. Shoulder electromyography in multidirectional instability. J Shoulder Elbow Surg 2004;13:24–9.
[41] von Eisenhart-Rothe R, Mayr HO, Hinterwimmer S, Graichen H. Simultaneous 3D assessment of glenohumeral shape, humeral head centering, and scapular positioning in atraumatic shoulder instability: a magnetic resonance-based in vivo analysis. Am J Sports Med 2010;38:375–82.
[42] von Eisenhart-Rothe R, Matsen FA, Eckstein F, Vogl T, Graichen H. Pathomechanics in atraumatic shoulder instability - Scapular positioning correlates with humeral head centering. Clin Orthop Relat Res 2005;433:82–9.
[43] Maki NJ. Cineradiographic studies with shoulder instabilities. Am J Sports Med 1988;16:362–4.
[44] Armfield DR, Stickle RL, Robertson DD, Towers JD, Debski RE. Biomechanical basis of common shoulder problems. Semin Musculoskelet Radiol 2003;7:5–18.
[45] Sgroi TA, Cilenti M. Rotator cuff repair: post-operative rehabilitation concepts. Curr Rev Musculosk Med 2018;11:86–91.
[46] Kronberg M, Brostrom LA, Nemeth G. Differences in shoulder muscle activity between patients with generalized joint laxity and normal controls. Clin Orthop Relat Res 1991;269:181–92.
[47] Ludewig PM, Reynolds JF. The association of scapular kinematics and glenohumeral joint pathologies. J Orthop Sports Phys Ther 2009;39:90–104.
[48] Itoi E, Motzkin NE, Morrey BF, An KN. Scapular inclination and inferior stability of the shoulder. J Shoulder Elbow Surg 1992;1:131–9.
[49] Basmajian JV, Bazant FJ. Factors preventing downward dislocation of the adducted shoulder joint. An electromyographic and morphological study. J Bone Joint Surg Am 1959;41-a:1182–6.
[50] Karduna AR, McClure PW, Michener LA, Sennett B. Dynamic measurements of three-dimensional scapular kinematics: a validation study. J Biomech Eng 2001;123:184–90.
[51] Pizzari T, Wickham J, Balster S, Ganderton C, Watson L. Modifying a shrug exercise can facilitate the upward rotator muscles of the scapula. Clin Biomech 2014;29:201–5.
[52] Warner JJ, Micheli LJ, Arslanian LE, Kennedy J, Kennedy R. Scapulothoracic motion in normal shoulders and shoulders with glenohumeral instability and impingement syndrome. A study using Moire topographic analysis. Clin Orthop Relat Res 1992;285:191–9.
[53] Jaggi A, Lambert S. Rehabilitation for shoulder instability. Br J Sports Med 2010;44:333–40.
[54] Merolla G, Cerciello S, Chillemi C, Paladini P, De Santis E, Porcellini G. Multidirectional instability of the shoulder: biomechanics, clinical presentation, and treatment strategies. Eur J Orthop Surg Traumatol 2015;25:975–85.
[55] Ettinger L, Weiss J, Shapiro M, Karduna A. Normalization to Maximal Voluntary Contraction is Influenced by Subacromial Pain. J Appl Biomech 2016;32:433–40.
[56] Cid MM, Januario LB, Zanca GG, Mattiello SM, Oliveira AB. Normalization of the trapezius sEMG signal - a reliability study on women with and without neck-shoulder pain. Brazilian J Phys Therap 2017.
[57] Pataky TC. Generalized n-dimensional biomechanical field analysis using statistical parametric mapping. J Biomech 2010;43:1976–82.